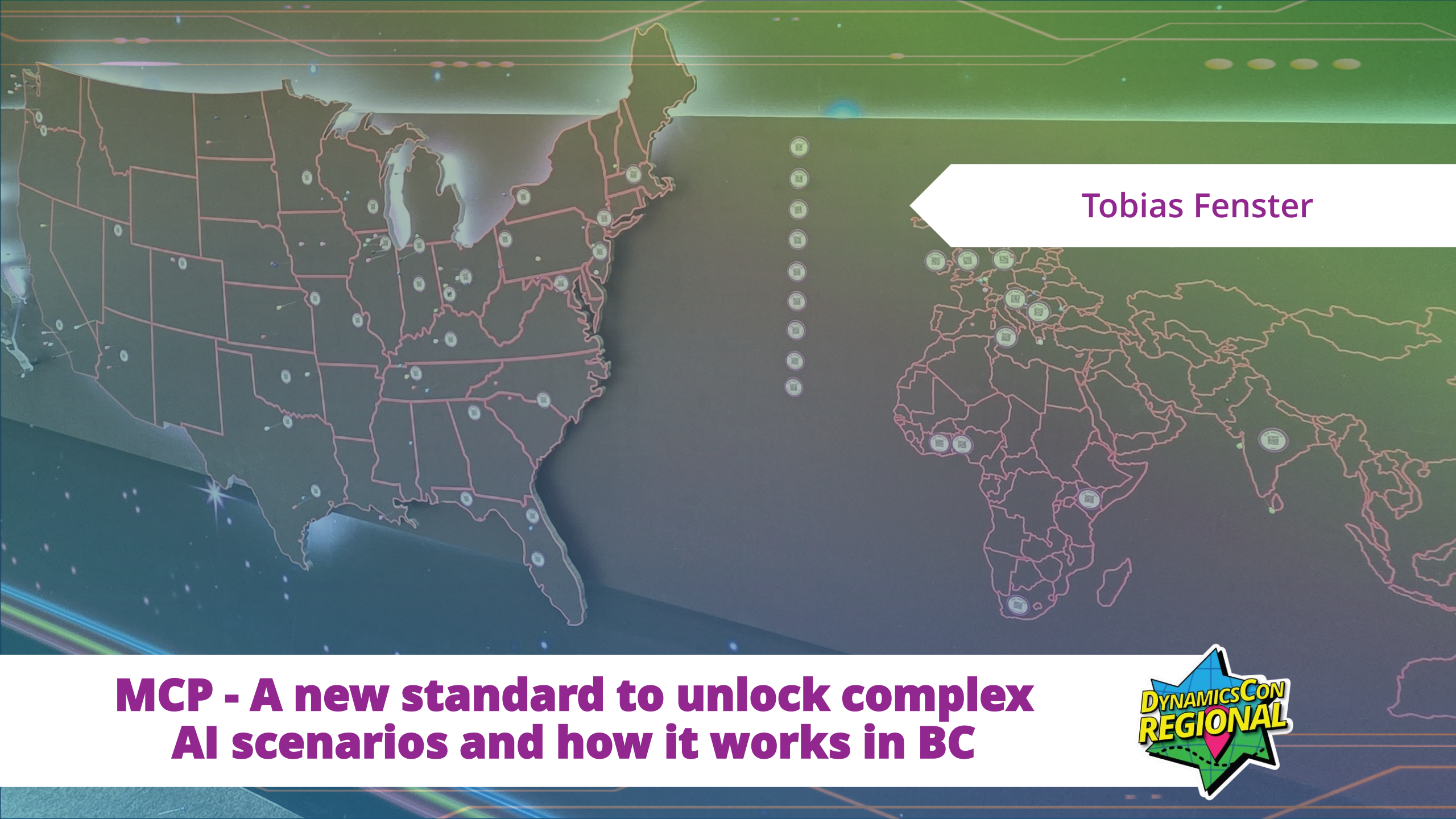




Tobias Fenster

**MCP - A new standard to unlock complex
AI scenarios and how it works in BC**





Speaker intro

Business

Managing Director bei 4PS DE und Chief Engineer bei Hilti

Community

MS Regional Director und MVP für Azure und BizApps

Docker Captain

Online

LinkedIn: [linkedin.com/in/tobiasfenster](https://www.linkedin.com/in/tobiasfenster)

Blog & Bluesky: tobiasfenster.io

Podcast: „Window on Technology“





Agenda

- Overview of MCP in general
- Overview of MCP in Business Central
- Advanced scenarios

All with a ton of demos 😊





Disclaimer

- Some of the content is from Microsoft (Jens Møller-Pedersen, Kennie Pontoppidan), shared with their consent
- My content has the DynamicsCon design, Microsoft's content has their design without the DynamicsCon logos





Introduction

- What is the Model Context Protocol (MCP)?
Open, standardized interface for LLMs to interact with external tools, data sources and APIs.
<https://modelcontextprotocol.io/>
- Why do we need it?
Models know their training data, but nothing else → MCP allows them to “plug in” external additions in a unified way, therefore the USB comparison
- Primarily created and maintained by Anthropic, but adopted by a lot of AI players





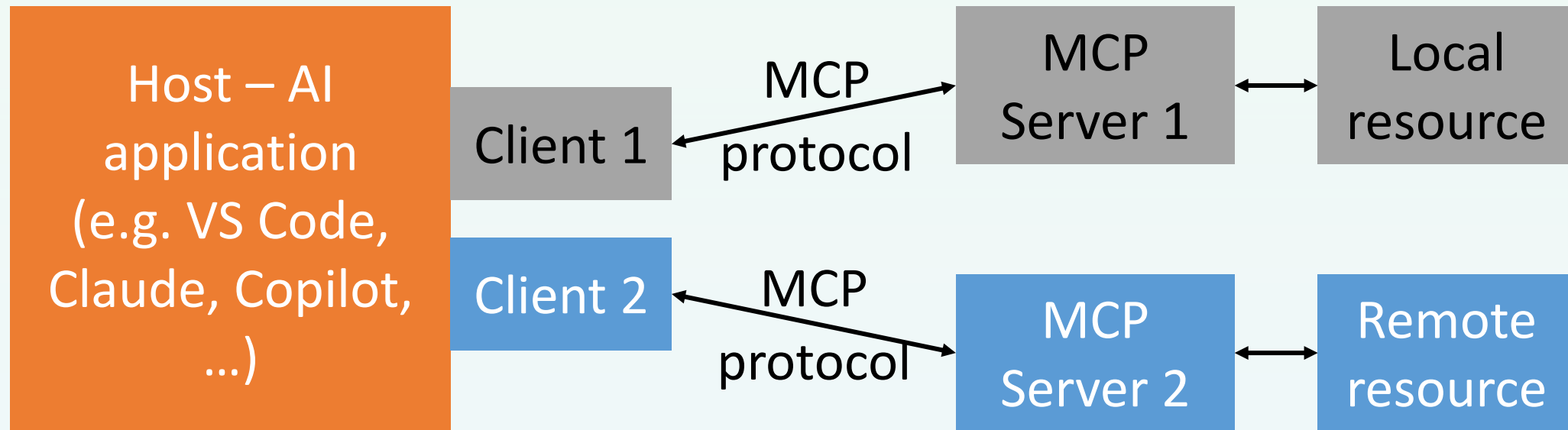
General Demos with VS Code

- Easier to see what happens
- Would also work with Claude, Copilot, Copilot Studio, ...





High-level architecture





Core concepts

- Tools:

Server exposes tools to be invoked by LLMs, enabling them to talk to external systems like databases and APIs or to perform tasks





Core concepts

- Tools:
List

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "tools/list",
  "params": {
    "cursor": "optional-cursor-value"
  }
}
```

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "tools": [
      {
        "name": "get_weather",
        "title": "Weather Information Provider",
        "description": "Get current weather information for a location",
        "inputSchema": {
          "type": "object",
          "properties": {
            "location": {
              "type": "string",
              "description": "City name or zip code"
            }
          },
          "required": ["location"]
        }
      }
    ],
    "nextCursor": "next-page-cursor"
  }
}
```





Core concepts

- Tools:
Call

```
{
  "jsonrpc": "2.0",
  "id": 2,
  "method": "tools/call",
  "params": {
    "name": "get_weather",
    "arguments": {
      "location": "New York"
    }
  }
}
```

```
{
  "jsonrpc": "2.0",
  "id": 2,
  "result": {
    "content": [
      {
        "type": "text",
        "text": "Current weather in New York:\n 72°F Partly cloudy"
      }
    ],
    "isError": false
  }
}
```





Core concepts

- Resources:
Server exposes resources as context for LLMs like files or database schemas





Core concepts

- Resources:

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "resources": [
      {
        "uri": "file:///project/src/main.rs",
        "name": "main.rs",
        "title": "Rust Software Application Main File",
        "description": "Primary application entry point",
        "mimeType": "text/x-rust"
      }
    ],
    "nextCursor": "next-page-cursor"
  }
}
```





Core concepts

- Prompts:
Server exposes prompt templates to users as instructions for interacting with language models, allowing clients to provide prompt lists and customize them when calling
- Sampling:
Server asks client to generate something via an LLM → Server can behave as an AI agent, but client is in control of model, authentication, sensitive data etc
- Elicitation:
Server asks client to request information from user → Server can request information dynamically, but client is in control of interaction and data sharing





MCP Inspector

MCP Inspector v0.21.1

Transport Type

Streamable HTTP

URL

http://localhost:8080/mcp

Connection Type

Via Proxy

Server Entry Servers File

Authentication

Configuration

Reconnect Disconnect

Connected

Docker AI MCP Gateway

Version: 2.0.1

ResourcesPromptsToolsTasksAppsPingSamplingElicitationsRootsAuthMetadata

Tools

List Tools

Clear

convert_time

Convert time between timezones

get_current_time

Get current time in a specific timezones

get_current_time

Get current time in a specific timezones

X Read-onlyDestructiveX IdempotentOpen-world

timezone *

IANA timezone name (e.g., 'America/New_York', 'Europe/London'). Use 'UTC' as local timezone if no timezone provided by the user.

Tool-specific Metadata:

No metadata pairs.

Add Pair

Run Tool

Copy Input





Demo

Using MCP Servers from VS Code



Microsoft



MCP

MCP Server vision

Make Business Central a first-class participant in AI workflows through **secure, reliable** MCP tools that agents and orchestrators can seamlessly use and compose.

Make all functionality available to AI and agents.



Demo

Interacting with BC through the MCP Server



The default MCP configuration

Default configuration

No API pages explicitly added
Allows dynamic addition of tools

Tools available



Dynamics 365 Business Ce...

In progress



Model Context Protocol **Initialized**

Description

Provides MCP Server access to Dynamics 365 Business Central (Preview).

Tools

Tool name	Description
bc_actions_search	Search for available Business Central API actions by keywords or natural language query. Use with 'bc_actions_describe' then 'bc_actions_invoke'.
bc_actions_describe	Get the detailed argument schema for a specific Business Central action (from 'bc_actions_search').
bc_actions_invoke	Invoke a Business Central action using arguments, that match the schema from 'bc_actions_describe'.

Dynamic Tools read flow explained

MCP Client



Pass keywords to
bc_actions_search



Call **bc_actions_describe**
for further info



Call **bc_actions_invoke**
passing filters

Business Central



Semantic search to find
top X relevant tools



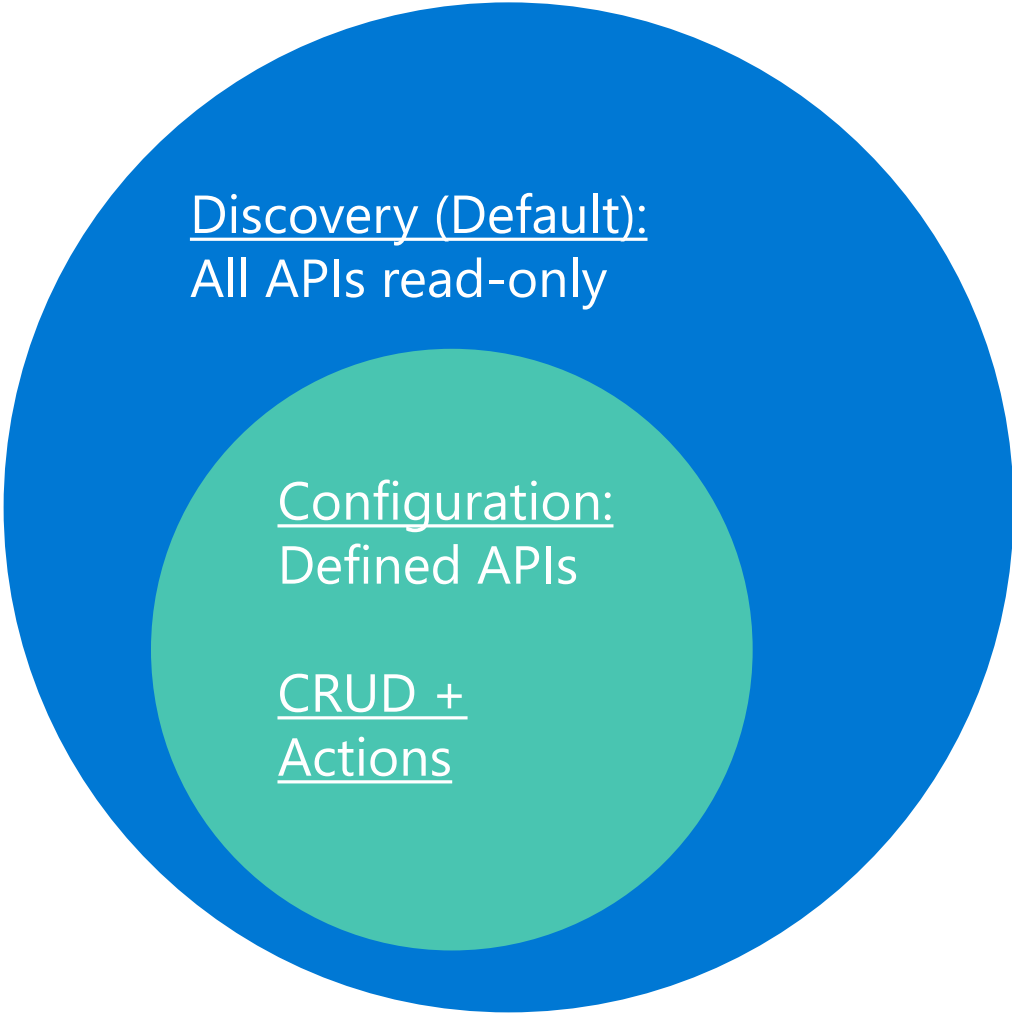
Return full description +
fields



Return data from API page
through OData* protocol

* Subject to changes

Introducing MCP configurations



Discovery (Default):
All APIs read-only

Configuration:
Defined APIs

CRUD +
Actions

MCP configurations limit the surface area

All MCP clients should specify a configuration, or fallback to the default

By default, only read operations are exposed



Demo

Defining and using MCP Server configurations



Business Central MCP Server: Modes

Static Tool Mode

For when you create dedicated agents. Useful for multi-agent use-cases.

Great, when the number of tools / APIs is small and manageable.

Dynamic Tool Mode

Exploratory agents. These agents can use search to find the relevant tools and self-adjust.

Great, when the available number of APIs is large.

Two dynamic tool modes:

- Tools in configuration
- All tools (read-only)



Improving Tools

- Tool descriptions and where they come from
- A practical approach
- Supporting all Microsoft APIs and API queries (since 28.2)
- Query tools (since 29)



Automatically generated tools descriptions for API pages

ListJobQueueLogEntries_PAG30090

List records for an entry JobQueueLogEntries
<AboutText for API page>

Instructions on OData constructs (e.g. \$filter, \$select etc...)

Available properties:

FieldName1 [type]

FieldName2 [type]

Add AboutTitle and AboutText to your API objects

The screenshot displays the SAP S/4HANA General Ledger interface. On the left, a sidebar contains navigation links: Home, Posting, General, Tax, and Balance Sheet. Below these are sections for 'General', 'Dimensions' (with codes for Global and Shortcut dimensions), 'Currency', 'Background Posting', and 'Reporting'. The main area is titled 'Tell Me' and has tabs for 'Advanced' and 'Preview'. A search bar contains the query 'how can I analyze fixed asset depreciations?'. Below the search bar, results are categorized into 'Go to Pages and Tasks' and 'Go to Reports and Analysis'. The 'Go to Reports and Analysis' section highlights the 'Fixed Asset Analysis' report, which is described as a flexible reporting option. A tooltip provides further details about this report.

General Ledger

Home Posting General Tax Ba

Change Payment Tolerance

General >

Dimensions

Global Dimension 1 Code
Global Dimension 2 Code
Shortcut Dimension 1 Code
Shortcut Dimension 2 Code
Shortcut Dimension 3 Code

Currency >

Background Posting >

Reporting >

Tell Me **Advanced** Preview

This feature uses AI to find pages and reports. [Read more](#)

🔍 how can I analyze fixed asset depreciations?

Go to Pages and Tasks Show all (15)

- > Calculate Depreciation Tasks
- > Index Fixed Assets Tasks
- > Depreciation Books Administration
Here you overview all registered depreciation books that you use in the Fixed Asset c...

Go to Reports and Analysis Show all (21)

- 📄 Fixed Asset Analysis Reports and Analysis
Fixed Asset Analysis Report is a flexible reporting option that provides a comprehensi...
- 📄 Fixed Asset Analysis (Excel)
Get an overview of your fixed assets for a gi
- 📄 Fixed Asset Maintenance Analysis

Search for 'how can I analyze fixed asset depre

- 🔍 Search company data
- 🔍 Search Help

Fixed Asset Analysis Report is a flexible reporting option that provides a comprehensive examination of an organization's fixed assets, such as property, plant, and equipment (PP&E), for different purposes. If the purpose is to reconcile asset values with GL then fields like acquisition, depreciation can be selected. If the purpose is about reviewing net value along with write down value, users can use this report accordingly by choosing the relevant amount fields and amount fields for multiple periods.



A practical approach

- Let AI generate
- Check manually

```
namespace Fps.BaseAPI.V16.Hr;

using Microsoft.HumanResources.Absence;

page 11362520 "APICauseOfAbsenceV164PS"
{
    PageType = API;
    APIPublisher = '4ps';
    APIGroup = 'base';
    APIVersion = 'v1.6';
    EntityName = 'causeOfAbsence';
    EntityCaption = 'Cause Of Absence';
    EntitySetName = 'causeOfAbsences';
    EntitySetCaption = 'Cause Of Absences';
    AboutText = 'Manages Cause Of Absences data. Supports CRUD operations (GET, POST, PATCH, DELETE) for integrat
    SourceTable = "Cause of Absence";
    DelayedInsert = true;
    ODataKeyFields = SystemId;

    layout
    {
        area(Content)
        {
            repeater(GroupName)
            {
                field(code; Rec.Code) { Caption = 'Code'; }
                field(description; Rec.Description) { Caption = 'Description'; }
                field(unitOfMeasureCode; Rec."Unit of Measure Code") { Caption = 'Unit Of Measure Code'; }
                field(causeOfAbsenceCostCode; Rec."Cost Code") { Caption = 'Cause Of Absence Cost Code'; }
                field(colourInGanttChart; Rec."Colour in Gantt Chart") { Caption = 'Colour In Gantt Chart'; }
            }
        }
    }
}
```



Business Central MCP Server: Modes

Additional things to consider

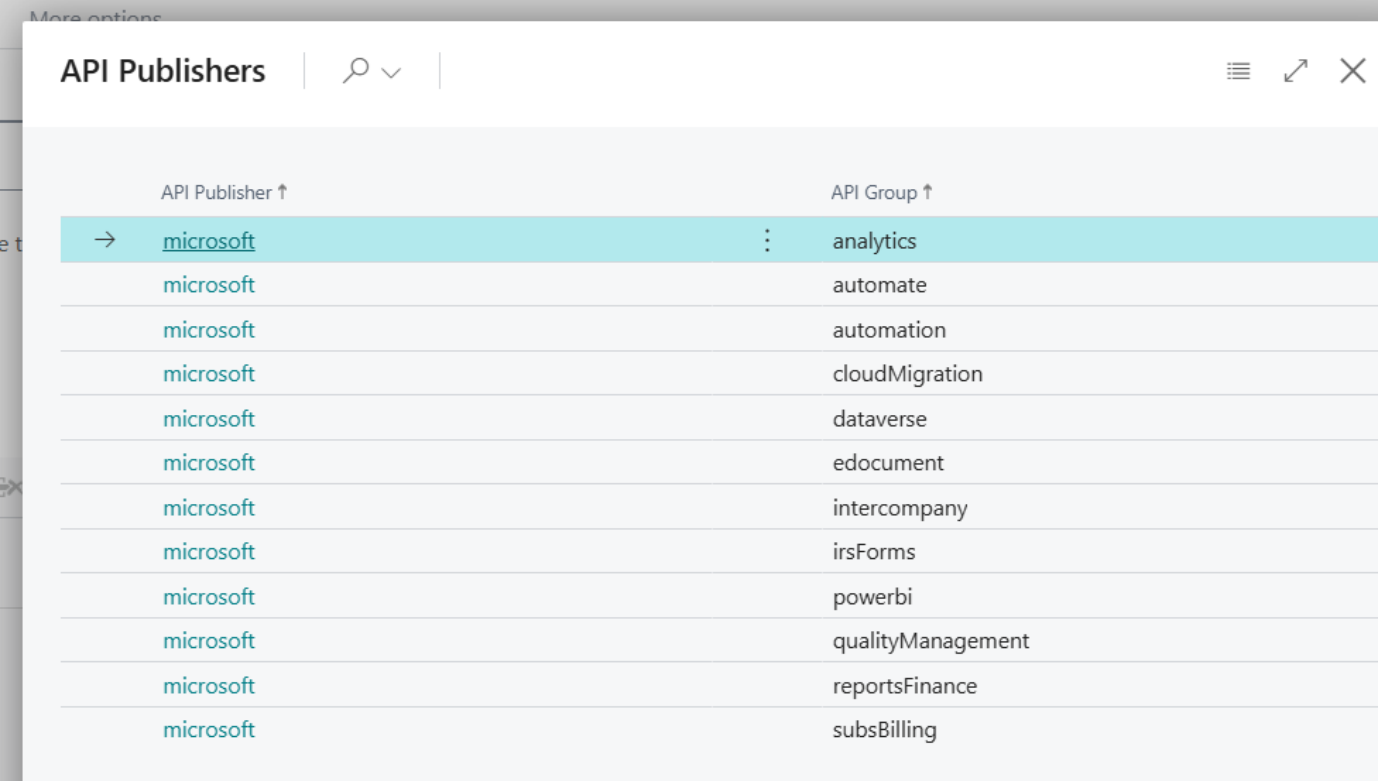
Each tool is loaded into the model's context window, filling it even when not in use. This leads to:

- Tool confusion for the LLM, along with excessive expansion of the context.
- Model performance decreases drastically as context increases. They make mistakes, become forgetful etc.

Different client have different limits on max number of tools supported.

Support for all Microsoft APIs

Now available in MCP
Server



The screenshot shows a web interface titled 'API Publishers'. It features a search bar and a list of API publishers. The first publisher, 'microsoft', is selected and expanded, showing a list of API groups. The interface is clean and modern, with a light blue and white color scheme.

API Publisher ↑	API Group ↑
→ microsoft	analytics
microsoft	automate
microsoft	automation
microsoft	cloudMigration
microsoft	dataverse
microsoft	edocument
microsoft	intercompany
microsoft	irsForms
microsoft	powerbi
microsoft	qualityManagement
microsoft	reportsFinance
microsoft	subsBilling

Deluxe agent configuration

 Copy

 Validate

Advanced

More options

General >

In Static Tool Mode, objects in the available tools will be directly exposed to clients. You can manage these tools by adding, modifying, or removing them from the configuration.

Add Tools by API Group

API Publisher	microsoft	...
---------------------	-----------	-----

API Group

OK

Cancel

Allow
Create

Allow
Modify

Allow
Delete

Allow
Bound
Actions

 Copy Validate **Advanced** More options

 New Line

→ Page

×

Cancel



✓ Saved



Deluxe agent configuration

Copy Validate Advanced | More options

General >

In Static Tool Mode, objects in the available tools will be directly exposed to clients. You can manage these tools by adding, modifying, or removing them from the configuration.



Available Tools

New Line Delete Line Select Tools... Add Tools by API Group Add All Standard APIs as Tools



API Object Type ↑			Object ID ↑	Object Name	API Version	Allow Read	Allow Create	Allow Modify	Allow Delete	Allow Bound Actions
→	Page	⋮	20766	APIV2 Expanded Permission Sets	v2.0	☒	☐	☐	☐	☐
	Page		20767	APIV2 - Access Control	v2.0	☒	☐	☐	☐	☐
	Page		20768	APIV2 - Users Permissions	v2.0	☒	☐	☐	☐	☐
	Page		20769	APIV2 - User Permission Sets	v2.0	☒	☐	☐	☐	☐
	Page		20770	APIV2 - Permission Sets	v2.0	☒	☐	☐	☐	☐
	Page		30000	APIV2 - Aut. Config. Packages	v2.0	☒	☐	☐	☐	☐
	Page		30001	APIV2 - Aut. Companies	v2.0	☒	☐	☐	☐	☐
	Page		30002	APIV2 - Aut. Extensions	v2.0	☒	☐	☐	☐	☐
	Page		30004	APIV2 - Aut. Users	v2.0	☒	☐	☐	☐	☐
	Page		30005	APIV2 - Aut. Permission Sets	v2.0	☒	☐	☐	☐	☐
	Page		30006	APIV2 - Aut. Extension Upload	v2.0	☒	☐	☐	☐	☐

Support for API queries

API queries are the foundation for our Power BI apps

Now this is available in MCP Server

MCP Query API Tools						
ID ▼	Name ↑	Entity Name ▼	API Publisher	API Group	API Vers	
6212	Employee Absence - PBI API	pbiEmployeeAbsence	microsoft	analytics	v0.5,v1	
6214	Employee Quali - PBI API	pbiEmployeeQualification	microsoft	analytics	v0.5,v1	
6213	EmployeeLedgerEntry - PB...	pbiEmployeeLedgerEntry	microsoft	analytics	v0.5,v1	
6215	Employees - PBI API	pbiEmployee	microsoft	analytics	v0.5,v1	
36958	G/L Account Categories	generalLedgerAccountCategory	microsoft	analytics	v0.5,v1	
36959	G/L Accounts	generalLedgerAccount	microsoft	analytics	v0.5,v1	
36960	G/L Budget Entries - PBI API	generalLedgerBudgetEntry	microsoft	analytics	v0.5,v1	
36961	G/L Budgets	generalLedgerBudget	microsoft	analytics	v0.5,v1	
36956	G/L Entries - Closing	closingGeneralLedgerEntry	microsoft	analytics	v0.5,v1	
36962	G/L Entries - Income State...	incomeStatementGeneralLedgerEntry	microsoft	analytics	v0.5,v1	
37014	G/L Income Statement - P...	incomeStmtGeneralLedgerEntry	microsoft	analytics	v0.5,v1	
36955	G\L Entries - Balance Sheet	balanceSheetGeneralLedgerEntry	microsoft	analytics	v0.5,v1	
37020	Inv. Adj. Ent Order - PBI API	inventoryAdjustmentEntryOrder	microsoft	analytics	v0.5,v1	
36999	Item Budget Entries - Purch.	purchaseItemBudgetEntry	microsoft	analytics	v0.5,v1	
37004	Item Budget Entries - Sales	salesItemBudgetEntry	microsoft	analytics	v0.5,v1	
37002	Item Budget Names	itemBudget	microsoft	analytics	v0.5,v1	
36968	Item Ledger Entries - PBI ...	itemLedgerEntry	microsoft	analytics	v0.5,v1	
36986	Item Ledger Entries - Prod.	prodItemLedgerEntry	microsoft	analytics	v0.5,v1	
36953	Items - PBI API	item	microsoft	analytics	v0.5,v1	
36992	Job Ledger Entries - PBI API	jobLedgerEntry	microsoft	analytics	v0.5,v1	

MCP Query API Tools



ID ▼	Name ↑	Entity Name ▼	API Publisher	API Group	API Version
6212	Employee Absence - PBI API	pbiEmployeeAbsence	microsoft	analytics	v0.5,v1.0
6214	Employee Quali - PBI API	pbiEmployeeQualification	microsoft	analytics	v0.5,v1.0
6213	EmployeeLedgerEntry - PB...	pbiEmployeeLedgerEntry	microsoft	analytics	v0.5,v1.0
6215	Employees - PBI API	pbiEmployee	microsoft	analytics	v0.5,v1.0
36958	G/L Account Categories	generalLedgerAccountCategory	microsoft	analytics	v0.5,v1.0
36959	G/L Accounts	generalLedgerAccount	microsoft	analytics	v0.5,v1.0
36960	G/L Budget Entries - PBI API	generalLedgerBudgetEntry	microsoft	analytics	v0.5,v1.0
36961	G/L Budgets	generalLedgerBudget	microsoft	analytics	v0.5,v1.0
36956	G/L Entries - Closing	closingGeneralLedgerEntry	microsoft	analytics	v0.5,v1.0
36962	G/L Entries - Income State...	incomeStatementGeneralLedgerEntry	microsoft	analytics	v0.5,v1.0
37014	G/L Income Statement - P...	incomeStmtGeneralLedgerEntry	microsoft	analytics	v0.5,v1.0
36955	G\L Entries - Balance Sheet	balanceSheetGeneralLedgerEntry	microsoft	analytics	v0.5,v1.0
37020	Inv. Adj. Ent Order - PBI API	inventoryAdjustmentEntryOrder	microsoft	analytics	v0.5,v1.0
36999	Item Budget Entries - Purch.	purchaseItemBudgetEntry	microsoft	analytics	v0.5,v1.0
37004	Item Budget Entries - Sales	salesItemBudgetEntry	microsoft	analytics	v0.5,v1.0
37002	Item Budget Names	itemBudget	microsoft	analytics	v0.5,v1.0
36968	Item Ledger Entries - PBI ...	itemLedgerEntry	microsoft	analytics	v0.5,v1.0
36986	Item Ledger Entries - Prod.	prodItemLedgerEntry	microsoft	analytics	v0.5,v1.0
36953	Items - PBI API	item	microsoft	analytics	v0.5,v1.0
36992	Job Ledger Entries - PBI API	jobLedgerEntry	microsoft	analytics	v0.5,v1.0

OK

Query tool for MCP Server

Have your MCP host construct queries.
Then we will compile and run them.
And get you the data.

No APIs needed.

New System tools

System Tools ▾

Tool Name ↑		Tool Description
<u>bc_data_find_tables</u>	⋮	Find Business Central tables by name o...
bc_data_get_table_relations		Get outgoing or incoming relations for...
bc_data_get_table_schema		Get the fields for a Business Central ta...
bc_data_query		Compile and validate an AL query. By d...



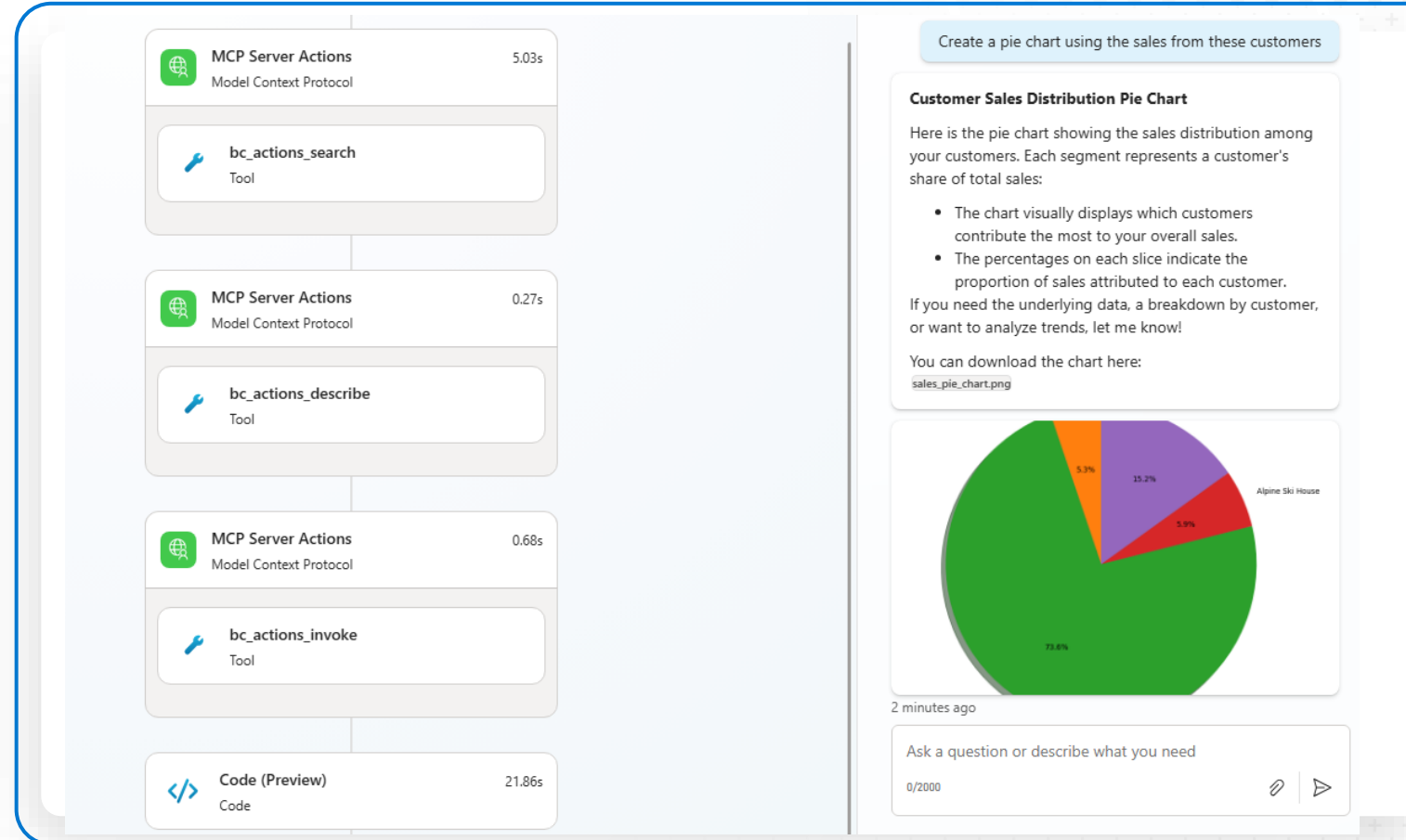
Demo

API groups, queries, and query tools



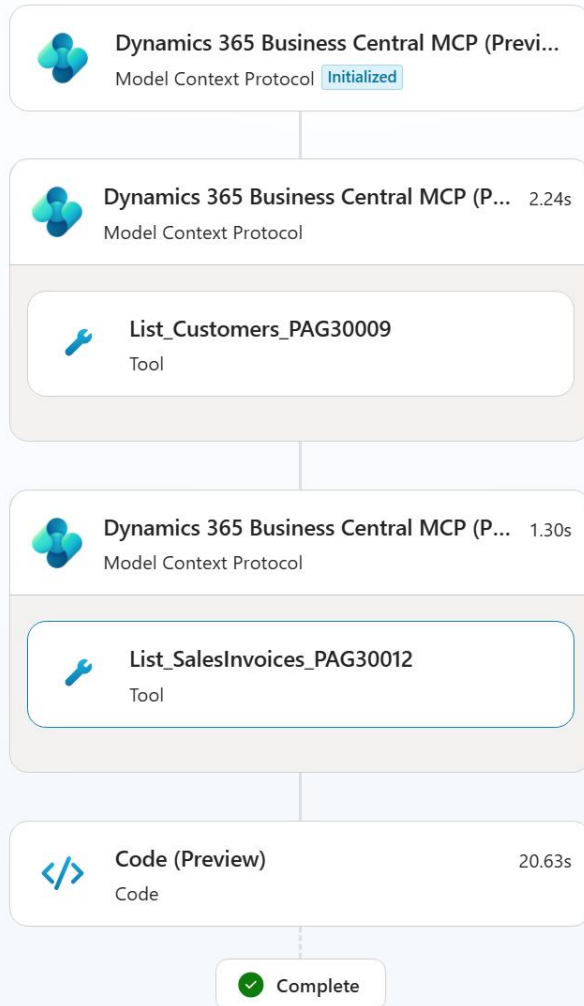
Embedded Resources to enable code-execution

- Data is returned in a structured format
- Handle larger amount of data with better context management.
- Use code-execution tools for Analysis



Example

Test your agent



Dynamics 365 Business Central MCP (Previous version) (Model Context Protocol) Complete

```
Content (table)
3 |   "text": "Returned 100 of 303 total
4 |   records. Use 'top' and 'skip' in
5 |   'RequestParameters' to paginate.",
6 |   "type": "text"
7 | }
8 | {
9 |   "file": {
10 |     "$kind": "FileDataValue",
11 |     "value": {
12 |       "$kind": "ConversationFileReference",
13 |       "value":
14 |         "93c9a542-d02a-43a2-9298-6e7c59dce70c"
15 |     }
16 |   }
17 | }
```

Copy

> Raw output

Rationale ⓘ

[Show rationale](#)

Edit configuration Documentation Feedback

New test session Evaluate Variables ... Fullscreen Close

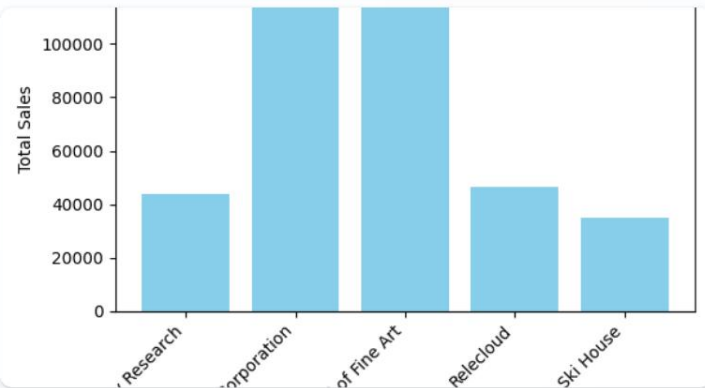
Show me the sales distribution of my customers in a bar chart

Sales Distribution of Customers

Here is the bar chart showing the total sales per customer:

- **Adatum Corporation:** 130,272.18
- **School of Fine Art:** 119,475.89
- **Trey Research:** 43,768.79
- **Relecloud:** 46,409.13
- **Alpine Ski House:** 35,025.91

The chart visually represents these totals, with customer names on the x-axis and total sales on the y-axis. If you need further breakdowns or analysis, let me know!



20 minutes ago

Ask a question or describe what you need

0/2000



You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#)

Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Example

Test your agent



Dynamics 365 Business Central MCP (Previ...

Model Context Protocol Initialized



Dynamics 365 Business Central MCP (P... 2.24s

Model Context Protocol



List_Customers_PAG30009

Tool



Dynamics 365 Business Central MCP (P... 1.30s

Model Context Protocol



List_SalesInvoices_PAG30012

Tool



Code (Preview)

Code

20.63s



Complete



Code (Preview)

Complete

Code

corporation: 130,272.18 School of Fine Art: 119,475.89
46409.13 Alpine Ski House: 35025.91 [1] [2]

Python

```
connector_client: ConnectorClient, input: Dict[str, Any]) -> ExecutionResult:
    output = {"text": "", "error": {"message": ""},
            "files": [], "sourceFiles": []}
    message_id = input.get("RequestId", "")
    # File IDs for processing
    customer_file_id = "4d69f71c-437a-43d4-b221-d8bfdafea1b0"
    invoice_file_id = "93c9a542-d02a-43a2-9298-6e7c59dce70c"
    file_names = [customer_file_id, invoice_file_id]
    logger.info("Reading input files: %s", file_names)

    files_dict = read_multiple_files_from_input(input,
                                                message_id, file_names)
    # Track which files were read
    for fn in file_names:
        if fn in files_dict:
            output["sourceFiles"].append(fn)
```

Rationale ⓘ

Show rationale

+ New test session 🔄 Evaluate {x} Variables ... 📄 ✕

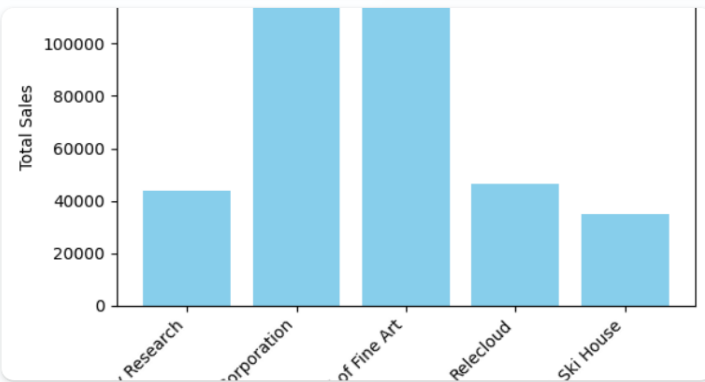
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The chart visually represents these totals, with customer names on the x-axis and total sales on the y-axis. If you need further breakdowns or analysis, let me know!



2 minutes ago

Ask a question or describe what you need

0/2000



You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#)
Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Embedded Resources

With

- Handle larger amounts of data.
- MCP returns a file-reference which can be passed to the code-execution tools for Advanced Analysis.

Without

- All data returned, is added to the model's context.
- Data is returned in a "Text-Content" block and is directly ingested.



Wait, there's more (for admins)

- Admin MCP Server
- Query and manage environments
- Different URL
- Not configurable





Demo

Admin MCP Server





Wait, there's more (for devs)

- AL MCP Server
- For developer scenarios like compiling, publishing, testing etc.
- Out of scope for today 😊



Want to learn more?

aka.ms/bcMCP

Find by title

MCP in Business Central overview

Configure Business Central MCP Server

AL MCP Server

Connect to MCP server with Copilot Studio

Connect to MCP server with Visual Studio Code

Connect to MCP server with GitHub Copilot CLI and other hosts

> Designing and coding agents (preview)

> Integration

> Deprecated features

> Business Central on-premises

Download PDF

Learn > Dynamics 365 > Business Central >

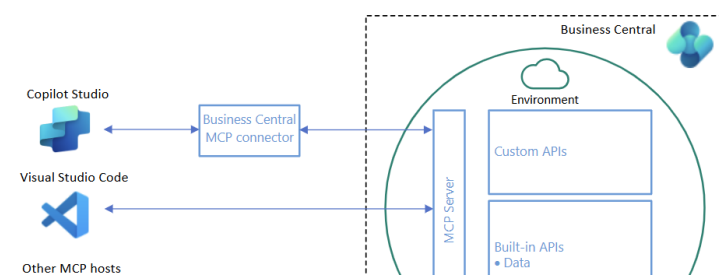
Ask Learn

Model Context Protocol (MCP) in Business Central overview

Summarize this article for me

APPLIES TO: Business Central online

The [Model Context Protocol \(MCP\)](#) is an open standard that defines how AI applications communicate with data sources and tools. It provides a consistent and secure way for AI clients—such as Copilot Studio, GitHub Copilot, Claude, ChatGPT, and custom agents—to access and interact with external systems like Business Central.



In this article

Business Central MCP Server

What the MCP Server can do

Supported MCP hosts

How MCP hosts connect to MCP server

Next steps

Related information

Was this page helpful?



<https://tobiasfenster.io>

Thanks! Questions?

